

3.3.3 AI Application Skeleton

Text prompt used

"Using our FamPlan design class diagram based on the Boundary-Control-Entity (BCE) pattern in TypeScript, generate a minimal skeleton folder hierarchy for the project. The structure must consist of empty files containing only // TODO comments indicating class responsibilities and method stubs. Do not write any implementation code, logic, mock returns, or library imports inside the files. Package the generated structure into a downloadable ai_skeleton.zip archive and output the corresponding directory tree in text."

AI Textual Output (Directory Tree Representation)

Please refer to ai_skeleton.zip uploaded in the github repository for the application skeleton

```
server/
├── boundaries/
│   ├── AccountBoundary.ts
│   ├── AdminMonitoringBoundary.ts
│   ├── DecisionSupportBoundary.ts
│   ├── DiscoveryBoundary.ts
│   ├── MapRouteBoundary.ts
│   ├── NeighbourhoodBoundary.ts
│   └── PreferenceBoundary.ts
├── controllers/
│   ├── AccountController.ts
│   ├── AdminMonitoringController.ts
│   ├── ComparisonController.ts
│   ├── DataSyncController.ts
│   ├── DiscoveryController.ts
│   ├── NeighbourhoodController.ts
│   ├── PreferenceController.ts
│   ├── RouteController.ts
│   ├── ScoringController.ts
│   ├── ShortlistController.ts
│   └── SyncScheduler.ts
├── entities/
│   ├── Facility.ts
│   ├── FamilyPreferenceProfile.ts
│   ├── FamilySuitabilityScore.ts
│   ├── NeighbourhoodProfile.ts
│   ├── SearchQuery.ts
│   ├── ShortlistEntry.ts
│   └── UserAccount.ts
└── external/
    ├── OneMapBoundary.ts
    └── PublicDataSourceBoundary.ts
```

Critical Evaluation of AI Output

- **Agreement with Choices:**

The AI followed our BCE class diagram accurately, cleanly separating classes into boundaries/, controllers/, entities/, and external/ directories.

- **Identified Omissions:**

It lumped all boundary classes into the server/ directory instead of separating client-side React UI boundaries from Express API routes. It also omitted shared TypeScript interfaces and database/ORM configuration (Prisma), which had to be added manually during prototype development.

- **Constraint Adherence:**

The AI strictly adhered to instructions. No executable logic, mock return statements, or library imports were generated; all files contain solely // TODO comments and stub signatures.